Alfred Navarro

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EDUCATION

University of Florida Gainesville, FL

B.S. Computer Engineering | GPA: 3.95/4.0 *Expected May 2028*

- **Coursework:** Data Structures, Programming, Computer Architecture (RISC-V), Computer Systems, Digital Logic (VHDL, FPGA), Discrete Structures, Linear Algebra, Physics, Calculus

EXPERIENCE

Undergraduate Research Assistant Aug 2026 – Present

UF Herbert Wertheim College of Engineering | Computational Research Group *Gainesville, FL*

- Research agentic AI under Prof. Ashish Aggarwal for Cipher, UF's AI learning assistant that tutors students through hints and questions instead of handing out answers
- Research a persistent memory layer that lets Cipher retain each student's progress, misconceptions, and past sessions to personalize tutoring over time
- Explore human-in-the-loop workflows that let instructors review, correct, and guide AI-generated responses before they reach students
- Study human-AI interaction in Socratic tutoring, focusing on how hint-based guidance affects student engagement and problem-solving

News Intern Jun 2024 – Aug 2024

JA Youth Employment / Island TV *Miami, FL*

- Completed a paid internship through Junior Achievement of South Florida, including weekly skill checks and reviews plus training in public speaking and interviewing
- Assisted on set at Island TV with lighting, camera, and equipment setup, and helped edit and create video content for the company's social media accounts

PROJECTS

CircuitEdge | *Arduino Learning Platform (Next.js 15, Supabase, Stripe, Cloudflare)* Mar 2026 – Present

- Developed and launched a paid full-stack SaaS that teaches Arduino from scratch, with 29 tutorials, 232 quiz questions, and about 14 hours of content
- Designed a Postgres schema (4 tables, 9 row-level security policies) and implemented OAuth-only sign-in (Google, GitHub) so the app never stores passwords
- Integrated $9/month subscriptions with Stripe checkout, a customer portal, and webhooks that keep billing status in sync with the database in real time
- Shipped 5 page routes and 5 API/auth endpoints covering the full flow from sign-in to checkout to content delivery

Embedded POS Terminal | *ESP32-S3, C++, Flask* Aug 2026 – Present

- Developed C++ firmware for a touchscreen payment terminal that reads NFC cards and talks to its display and reader over SPI and I2C
- Engineered the state machine that walks a sale from touch input to card tap to checkout, with a Flask server creating Stripe payments over Wi-Fi

Mosquito Tracker | *1st Place NASA geoEMERGE Hackathon (Astro, Svelte 5)* Jan 2026

- Engineered a lightweight version of NASA's GLOBE Observer app that loads 6x faster (0.5s LCP) with a 75x smaller payload and a perfect 100/100/100/100 Lighthouse score
- Built offline-first support (IndexedDB queueing, Service Worker background sync, on-device WebP compression), cutting upload sizes by 70–90%
- Integrated 3 NASA datasets (MODIS, GPM IMERG, Landsat 8/9) with the GLOBE Observer API through server-side proxy endpoints, and shipped to web, iOS, and Android

Emergency Resource Locator | *React, React-Leaflet, OpenStreetMap*

- Built the app's GUI and implemented real-time resource filters across multiple emergency categories on an interactive map
- Integrated React-Leaflet and OpenStreetMap APIs to render live, filterable location data, with fallbacks for unnamed locations and disaster-ready instructions

LockIN | *Embedded Study-Focus Device (Arduino Uno, C++)*

- Designed the circuitry and embedded C++ code for a phone-locking study device using a servo motor, IR remote, and LCD display, while teammates built the SolidWorks enclosure
- Programmed and debugged timer states, IR input processing, and servo-based lock actuation to enforce distraction-free study sessions

LEADERSHIP

Treasurer Feb 2026 – Present

UF Data Science & Informatics *Gainesville, FL*

- Manage an organization budget over $10,000, funding 8 tech workshops, 4 member-led projects, research initiatives, hackathons, and the annual Research Symposium
- Lead sponsorship coordinators in securing partnerships with StudyEdge, NLP Logix, VYSPAR, and NVIDIA through direct outreach
- Partner on site with the Artificial Intelligence and Informatics Research Institute (AIIRI) to coordinate budgeting and program planning
- Plan monthly career-readiness and data science events and promote general body meetings to drive attendance and member participation

Marketing Coordinator

IEEE Signal Processing Society (SPS) at UF *Gainesville, FL*

- Designed promotional campaigns in Canva, including posters, flyers, slides, and social media posts, that brought 50+ attendees to the chapter's first General Body Meeting
- Promote technical workshops in signal processing, turning topics like audio, image processing, and machine learning into clear visuals that appeal to students across engineering and computer science
- Work with chapter officers to plan event promotion and keep the society's branding consistent across platforms

Social Chair Jan 2026 – Present

Southern Scholarship Foundation *Gainesville, FL*

- Manage semester budgets of $250–$300, planning and executing fundraisers
- Promote foundation events through flyers, social media campaigns, and newsletters
- Design social events tailored to residents' interests to increase involvement

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, JavaScript, TypeScript, HTML/CSS, MATLAB, Assembly, Lua, LaTeX
Frameworks: React, Next.js, Node.js, Flask, FastAPI, Astro, Svelte, REST APIs
Tools: PyTorch, TensorFlow, Pandas, Supabase, Stripe, Cloudflare Pages, Git, VS Code, Visual Studio, Quartus Prime, Logisim-Evolution
Hardware: ESP32-S3, Arduino, FPGA, SPI, I2C, NFC
Certifications: NVIDIA Getting Started with Deep Learning, NVIDIA Fundamentals of Accelerated Data Science
Spoken Languages: English (fluent), Spanish (conversational)

HONORS AND AWARDS

1st Place, NASA geoEMERGE Hackathon (2026) • 3rd Place, UF Outreach Design Competition (2026) • HWCOE Dean's List (2025–2026) • AP Scholar with Distinction (2025) • Bright Futures Program Scholar (2025–Present)